



ARUN KUMAR CHAUBEY

ASSISTANT MANAGER(DATA SCIENTIST)

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PROFILE SUMMARY

With 6.8 years of experience in Data Science and Analytics Engineering, I have led strategic planning initiatives, enhanced data quality from 64% to 98%, and optimized processes to drive revenue growth. Skilled in SQL, Power BI, DAX, and Power Query, I have models (Regression, Decision Trees) to solve complex business problems. I have contributed to propensity modeling, fraud detection frameworks, and forecasting initiatives, enabling data-driven decision-making. Alongside this, I have spearheaded training programs, automated ETL workflows, and fostered a culture of continuous learning, driving performance improvements, operational efficiency, and measurable business success.

KEYSKILLS

- Dax Queries
- Power BI
- machine learning
- Classification
- AutoML (FLAML)
- Synthetic Data Generation
- Optuna
- PII Masking
- Python
- SSIS
- Predictive Modeling
- Clustering
- Hyperparameter Tuning (Optuna)
- Fraud Analytics
- Ollama
- SQL Server
- SQL
- Regression
- Statistical Data Analysis
- Explainable AI (SHAP)
- SHAP
- OCR (EasyOCR)

WORK EXPERIENCE

Assistant Manager(Data Scientist) - EXL

Oct 2024 - Present

EXL

- Developed comprehensive insurance fraud detection pipelines by integrating nine relational insurance datasets into a claim-level feature store containing 50,000 claims and over 60 engineered features, enhancing machine learning accuracy.
- Conducted exploratory data analysis (EDA), engineered predictive features, identified and mitigated target leakage, and optimized hyperparameters using Optuna to improve model performance.
- Applied AutoML model selection with FLAML, leveraging algorithms such as LightGBM, CatBoost, Random Forest, Decision Trees, and ensemble methods to construct robust fraud prediction models.
- Created explainable AI (XAI) workflows using SHAP to interpret model outcomes, providing global feature importance and claim-specific explanations for transparent fraud risk evaluation.
- Engineered diverse fraud detection features from claims, vehicles, claimants, policy history, liability, suspicious agents, MIAFTR, Capstone, and geographic risk datasets, emphasizing temporal, behavioral, and relational attributes.
- Automated data ingestion, ETL pipelines, feature engineering, and statistical analysis using Python (Pandas, NumPy, Scikit-learn), SQL, and SAS, improving data quality and accelerating model development cycles.
- Designed and developed interactive Power BI dashboards utilizing DAX, Power Query, and SQL to enable real-time monitoring of claims and underwriting risk metrics.

Manager (Data Scientist) - Nibav Lifts Private Ltd.

Jan 2023 - Dec 2023

Nibav Lifts Private Ltd.

- Developed predictive models using machine learning algorithms (Random Forest, Gradient Boosting, Regression, Decision Trees) to forecast sales, customer churn, and lead conversion, contributing to a 20% revenue increase.
- Built a demand forecasting model to optimize inventory and production planning, reducing delays and improving operational efficiency.
- Enhanced data quality from 64% to 98% through advanced analytics, ETL optimization (SSIS), and Python-based automation scripts.
- Created interactive dashboards and advanced analytics reports in Power BI, integrating DAX-driven KPIs and metrics for real-time business monitoring.
- Collaborated with cross-functional teams to identify actionable insights, enabling proactive strategic decisions in sales, operations, and

marketing.

Manager (Data Analyst & Scientist) - BYJUS

Feb 2020 - Jul 2022

BYJUS

- Built predictive models and statistical analyses (Regression, Decision Trees) to forecast sales performance, lead conversion, and customer engagement, improving conversion efficiency across multiple regions.
- Developed a student learning path recommendation model using behavioral data to suggest relevant courses and content, boosting student engagement and supporting upsell opportunities.
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- Conducted deep SQL-based sales funnel analysis to identify high-performing channels, optimize territory planning, and prioritize leads, leading to a 15% improvement in sales productivity.
- Designed optimized database schemas and automated ETL workflows (SSIS), enabling faster access to sales KPIs and reducing reporting delays by 30%.
- Delivered advanced Power BI dashboards, utilizing DAX and Power Query, to track funnel health, campaign ROI, subscription trends, and sales representative performance.
- Automated repetitive reporting and data preparation tasks using Python, freeing up 25% of analyst time for strategic initiatives.
- Collaborated with sales, product, and marketing teams to provide insights for targeted campaigns, pricing optimization, and course bundling strategies, directly contributing to revenue growth.

Executive Data Analyst - Extramarks Education

Jan 2019 - Feb 2020

Extramarks Education

- Leveraged SQL and Python for customer segmentation and predictive analytics, contributing to a 15% revenue uplift.
- Streamlined BI reporting using Power BI, cutting dashboard refresh times by 70% (from 15 minutes to 4.5 minutes).
- Collaborated internationally, optimizing data analytics workflows and increasing operational efficiency by 20%.

PROJECTS

Sales Analytics & AI-Driven Business Intelligence Platform

547 Days

- Led data engineering and business intelligence initiatives for NIBAV Home Lifts by designing and implementing scalable ETL pipelines and dynamic Power BI dashboards to track sales, revenue, and operational KPIs.
- Enhanced data quality from 64% to 98% using SSIS and SSIS transformations such as Join, Lookup, and Aggregator, ensuring accurate and dependable analytics and reporting.
- Developed interactive Power BI dashboards utilizing DAX and Power Query, delivering real-time insights and automated KPI monitoring for executive decision-making.
- Applied AI/ML methodologies, including Generalized Linear Models (GLM), to analyze business drivers, detect anomalies, and generate predictive insights that improved sales performance.

Training Program Development and Sales Performance Analysis

1033 Days

- Developed end-to-end ETL pipelines using SSIS and SQL Server, improving data quality from 64% to 98%.
- Built interactive Power BI dashboards using DAX and Power Query for sales, revenue, and geo-spatial analytics.
- Applied machine learning models (Linear Regression and Random Forest) to forecast sales trends and identify key business drivers supporting strategic planning.
- Performed EDA, feature engineering, and SQL-based analysis to uncover customer and regional sales patterns.
- Collaborated with stakeholders to translate analytical insights into business decisions, driving revenue growth.

Revenue & Efficiency Optimization Initiative at Extramarks

396 Days

- Led the Revenue & Efficiency Optimization Initiative at Extramarks, focusing on leveraging data analysis to drive business growth.
- Executed advanced SQL queries to extract, cleanse, and analyze customer data for actionable insights.
- Collaborated with global teams to resolve data inconsistencies and streamline operational workflows.
- Enhanced MS Power BI and Excel reporting dashboards, delivering clear, data-driven insights for strategic decision-making.
- Monitored key performance indicators and continuously refined reporting processes to improve accuracy and efficiency.
- Increased revenue by 15% through data-informed targeting and strategic initiatives.
- Boosted operational efficiency by reducing time on data-related tasks by 20%.
- Accelerated dashboard refresh rates from 15 to 4.5 minutes, enabling near real-time access to critical business metrics.

Fraud Detection & Claims Risk Analytics (UK Insurance Domain)

4 Months

- Built machine learning models (Regression, Decision Trees) to identify potentially fraudulent claims by analyzing claimant demographics, accident patterns, agent behavior, and historical fraud data.
- Integrated internal and external intelligence sources to create risk scores, reducing false positives and enabling early intervention in suspicious claims.
- Delivered an interactive fraud monitoring dashboard in Power BI with real-time KPIs for claims handlers and investigators.

Policy Conversion Propensity Model

3 Months

- Developed a propensity scoring model to predict the likelihood of converting policy submissions into active policies, using variables such as policy turnover, company age, broker success rate, premium size, claim history, and jurisdiction
- Performed feature engineering and applied statistical techniques (Regression, Decision Trees) to assign weighted scores.

Broker Relationship & Performance Scoring

2 Months

- Designed a broker scoring framework that considers submission volume, premium contribution, relationship history, and success rates across multiple years.
- Created percentile-based scoring logic to rank brokers and help prioritize high-value relationships.
- Built a broker insights Power BI dashboard integrating DAX measures, enabling relationship managers to identify growth opportunities.

Reporting & Business Intelligence

3 Months

- Delivered multiple Power BI dashboards for senior management to track KPIs across claims, fraud detection, underwriting conversion, and broker performance.
- Designed DAX measures and Power Query transformations to create real-time, actionable insights that drive strategic decision-making.

Synthetic Motor Insurance Fraud Data Generation & Feature Engineering

5 Months

- Designed and developed a production-ready synthetic motor insurance fraud data generation framework to enhance machine learning model development and analytics.
- Created realistic multi-table relational datasets simulating claims, vehicles, claimants, policy history, liability, suspicious agents, and external fraud sources.
- Modeled complex fraud scenarios including fraud rings, staged accidents, identity misrepresentation, and repeated entity reuse.
- Engineered aggregated and network-based fraud detection features to improve model accuracy.
- Injected real-world data quality issues and validated business rules to ensure dataset integrity.
- Produced privacy-compliant datasets optimized for fraud detection, preprocessing validation, and end-to-end machine learning pipeline testing.

PII Masking Pipeline for Insurance Claim Note

15 Days

- Utilized Python, Regex, Ollama, Qwen 2.5, Pandas, JSON, Multiprocessing, Excel, and Claude to develop a privacy-preserving PII masking pipeline for anonymizing insuranceclaim notes.
- Designed a hybrid solution combining deterministic regex-based masking for structured PII and locally hosted LLMs to extract unstructured entities such as names and addresses.
- Implemented a zero-hallucination architecture by generating structured JSON outputs with LLMs and performing precise string replacement in Python to ensure accurate, reliable masking.
- Built parallel processing with multi-worker execution, timeout handling, and review flags to efficiently manage large Excel datasets.
- Enabled secure data sharing for analytics and machine learning by protecting sensitive customer information through robust anonymization techniques.

Insurance Fraud Detection Pipeline

12 Months

- Developed an end-to-end insurance fraud detection pipeline by integrating 13 relational insurance datasets into a claim-level feature store containing 50,000 claims and 60+ engineered features.
- Engineered temporal, behavioral, and relational fraud detection features from claims, vehicles, claimants, policy history, liability, suspicious agents, MIAFTR, Capstone, and geographic risk data.
- Automated one-to-many data aggregation into claim-level features, including policy tenure, previous claims, claimant demographics, vehicle damage, and fraud linkage attributes.
- Identified and eliminated target leakage using permutation importance after detecting an unrealistic F1 score of 1.00 by removing post-investigation features.
- Retrained the fraud detection model using leakage-free features and selected CatBoost as the best-performing model through FLAML AutoML.
- Built an Explainable AI framework using SHAP to provide both global feature importance and claim-level prediction explanatio

EDUCATION

pranveer institute of technology, Kanpur

ADDITIONAL INFORMATION

Languages: English, Hindi

SOCIAL LINKS

[Linkedin.com/in/arun-k-chaubey](https://www.linkedin.com/in/arun-k-chaubey)